Resolvox

VoiceOps 2026 Capstone | Technical Report

What we built in the four required steps

Resolvox is a document-grounded voice support agent for ZentroMart. It handles policy questions, verified order follow-ups and confirmed human handover using the supplied synthetic dataset. The same support agent serves text, microphone and folder-based audio input.

Required step	Implemented behavior
1 - Answer with an LLM	An LLM API role/limits prompt and a prompt-only baseline without company documents or order tools. The shared assistant identifies itself as AI at the start of a conversation.
2 - Ground in documents	Policy sections retain IDs, authority, status, version and effective date. A local vector index and lexical search retrieve relevant original chunks. The LLM receives those chunks and source-derived facts; missing support leads to clarification or uncertainty.
3 - Hold a conversation	Session state preserves order context and follow-ups. The agent asks one missing-detail question, verifies access before private order disclosure, uses the supplied lookup_order and create_ticket tools, and confirms handover only from an actual tool result.
4 - Give it a voice	Gnani STT transcribes the customer clip, the same Step 3 agent responds, and Gnani TTS produces the spoken reply. English, Hindi and Hinglish are the primary supported languages.

Grounding and control of actions

Retrieval combines lexical matching with multilingual MiniLM embeddings in an ONNX int8 index and hybrid fusion. Current policy takes precedence over lower-authority material. Source-bound calculations handle return dates, business-day refund windows, payment-method units and membership conditions.

Operational authentication and tool confirmations are deterministic. When current policy requires an RRN, the host can append its exact authenticated event field; that record-reading step is identified in the trace. Business explanations are generated from actual evidence, with no practice-answer lookup or authored policy-answer fallback. Proof/contact fields are redacted from model input and public traces. Full generation chunks, model flags and tool outcomes are recorded separately from the spoken text.

Runtime

Implemented application profile: Groq openai/gpt-oss-20b with low reasoning effort, constrained JSON output and an additional source-claim review. Requests and repairs have bounded time budgets; no answer-provider fallback is enabled. Gnani is the only STT/TTS provider. Retrieval and the Streamlit interface run on the laptop. A separate local Ollama profile remains available.

Evaluation: Step 1 vs Step 2

Required comparison	Recorded score	Basis
Step 1 - Prompt-only LLM	1/52	Baseline artifact; 1 recorded error. No company documents or order tools.
Step 2 - Document-grounded answers	52/52	Stored actual practice-question predictions and targeted retests across recorded revisions.

The recorded baseline used qwen3:1.7b. The Step 2 total combines actual preserved outputs and targeted retests across earlier model/source revisions. Fresh selected checks now use the explicit Groq application profile. These configuration differences confound a causal comparison; a frozen same-model ablation is a remaining evaluation improvement.

Verified checks and recorded results

Selected check	Recorded result	Scope
Focused offline regression checks	432/432 verified	Return-date scope, source-support flags, repair context and release safeguards. Scripted clients; no live inference.
Refund and grounding regressions	655/655 verified	Source-bound dates, units, conditional eligibility, record reading and negative action/privacy checks.
Packaging and report regressions	148/148 verified	Evidence freshness, report limits, archive integrity and release guards.

Verified scope is limited to these overlapping offline selections; their counts are not additive. The latest live run contains source-review failures and an unavailable coupon reply, and the speech run did not pass all semantic checks. Groq daily-token quota limited further live verification. Original outputs and execution hashes remain preserved; no complete current live or speech pass is claimed.

These are exposed development regressions, not hidden organizer scores or a new full-suite run. The checks establish the stated selected behavior; they do not certify every unseen question or arbitrary semantic claim. Synthetic speech does not establish human accents, noisy-room performance or subjective listening quality.

Recorded selected generated-turn latency: 2.8-5.8 seconds across 7 saved turns. Host-only steps and speech service time are separate. These timings can include provider quota waits and are not a universal latency guarantee. Per-stage timings are preserved in JSON.

What went wrong and the fixes

Observed problem	Correction and verification
Missing placement date selected a personal return window	A source-bound guard rejects selecting the ordinary window because a placement date is unknown, including Hindi Unicode-hyphen phrasing. General rules and conditional exceptions remain valid, and requested fee evidence is retained during repair. Common Hinglish clothing nouns now bind to the catalog category.
Missing QC data became an unsupported status claim	A bound pending-status view distinguishes an unrecorded QC/initiation date from a failed check. An absent RRN cannot be offered for bank tracing. The source initiation clock remains anchored to pickup, with QC pass as a prerequisite. Earlier selected outputs and offline tests cover this distinction.
Wallet hours mixed with business days and an invented date	Known payment methods bind to the actual policy row. Wallet hour limits remain hours; other payment methods retain their business-day ranges and initiation anchor. Unsupported dated wallet estimates are rejected.
Compound refund questions lost requested coupon details	Coupon, partial-refund and amount requests retain the full original source/fact context through generation and repair. Earlier source-grounded outputs and targeted offline tests cover both requested facets.
Valid conditional/record wording was rejected	Source-backed conditional timing is separated from claims of performed actions. Historical completion uses a compact authenticated record-reading view instead of a future policy estimate. The actual recorded date and action/authentication boundaries were checked.
Hindi damage guidance abstained on remedy prerequisites	A source-only first reporting view explains the actual reporting window and required photos without promising an unverified benefit. Selected English, Hindi and Hinglish clips used real Gnani STT/TTS with synthetic input and preserved source hashes.

Interactive checks also exposed a credit-card phrase misread as a compensation-credit claim and a general paid-guarantee delivery request routed to order verification. The instrument scope and public-request routing were corrected. Targeted negative checks retain genuine credit conditions and private-record verification. Focused refund prompts now preserve the original question so requested membership facets remain visible.

Safeguards retained

Identity verification precedes private order disclosure. A policy fee cannot establish a personal paid/refund amount. Missing records do not prove a prohibition or failed event. Claim/quality verification is distinct from identity verification. Source metadata and full original validators remain active for focused generation; a rejected draft gets one bounded repair before uncertainty.

What we would improve next

Run a frozen same-model Step 1/Step 2 comparison and an independent unseen evaluation. Expand independent review of multi-part and qualitative claims: neither finite validators nor a model-based source review proves complete semantic entailment.

Measure hosted latency and throughput under deployment load and provider rate limits. Improve Hindi/Hinglish grammar and repetition, then test human speech with accents, background noise and unclear audio. Synthetic checks do not certify those conditions. Shared-customer deployment would additionally require authenticated private history and real store/ticket integrations.

Bonus work and reproducibility

Additional engineering beyond the basic build

Work	Practical benefit
Multilingual support	English, Hindi and Hinglish routing and spoken output share the same source-grounded agent. Only these primary routes are claimed here.
Local hybrid retrieval	Lexical/dense fusion, multilingual embeddings and ONNX int8 execution support document search on the laptop. Authority/effective-date metadata preserve policy precedence.
Source-bound safeguards	Checks cover recognized dates, units, amounts, prerequisites and unsupported action claims. Actual tool confirmations stay separate from generated explanations.
Inspectable evaluation and release	Saved input chunks, model flags, original execution epochs, audio hashes and machine-readable receipts make selected results reviewable. Retained outputs are labelled rather than presented as fresh inference.
Usable interface and batch processing	A responsive, single conversation supports Speak/Type, case details and supporting policies. Saved chats retain local text/audio/context; reopening does not restore private order verification. History databases stay outside Git and the archive. The batch runner saves text, JSON, real reply audio and source chunks.

Reproduce the agent

Code: https://github.com/Pranay207/Resolvox. Install the project dependencies, copy .env.example to the ignored .env, configure your own Gnani key in .env, provide the official extracted dataset and add your Groq key to the ignored .env.hosted for the configured application profile. The separate local profile uses Ollama qwen3:4b-instruct. Setup and environment details are in README.md.

Windows setup: powershell -ExecutionPolicy Bypass -File setup.ps1. Interactive profile: python scripts/run_app.py --hosted.

Hosted batch command: python run.py --hosted --audio_dir <in> --out_dir <out>. Omit --hosted only for the separately configured local Ollama profile. Each clip produces answer text, JSON, a real reply WAV and the source chunks supplied to generation; manifest.json records per-clip outcomes. Gnani remains the only speech service, including spoken recovery attempts.

Evidence accompanying the report

Step comparison: sample_outputs/eval_step1.json and eval_step2_latest_cases.json. Selected conversations/refunds: targeted_live_validation.json. Return-date check records: return_date_validation.json. Recorded speech: targeted_voice_validation.json. Current offline checks: release_update_verification.json. Earlier automated receipts retain their original source and test hashes. All paths are under sample_outputs/; their referenced artifacts retain their actual scopes.

Only the supplied synthetic customer data is used. Credentials, private input speech, virtual environments, dataset copies and model weights are excluded from the submission archive. This report describes implemented behavior and recorded results; organizer marks are determined by the separate hidden-audio evaluation.